Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for: **Paediatric Sepsis**	
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SUMMARY

Paediatric sepsis is a severe life-threatening organ dysfunction due to dysregulated host response to infection in children and young people under 18 years old.
This guideline is based on the NICE guideline 51 on *Sepsis: recognition, diagnosis and early management*, published in July 2016 and the *Surviving Sepsis Campaign* guidelines, published in October 2021
This guideline should be used in conjunction with the following Royal Devon University Healthcare NHS Foundation Trust guidelines:
Flow chart "*Could this be paediatric sepsis?*" (page 3);
Royal Devon Paediatric Early Warning Score (PEWS) charts;
https://hub.exe.nhs.uk/EasysiteWeb/getresource.axd?AssetID=4596&type=Full&servicetype=Attachment
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https://hub.exe.nhs.uk/EasysiteWeb/getresource.axd?AssetID=4599&type=Full&servicetype=Attachment
https://hub.exe.nhs.uk/EasysiteWeb/getresource.axd?AssetID=4598&type=Full&servicetype=Attachment
Royal Devon Antimicrobial Guidelines
Fever in Children aged 3 months to 5 years
Fever in Children under 3 months
Neutropenic Sepsis Guideline

KEY POINTS

Assessment for paediatric sepsis:

- Always consider "Could this be paediatric sepsis?" (see flow chart on page 3) in all unwell children, even if they do not have a high temperature
- The presentation of sepsis is often non-specific and not localised
- Pay particular attention to the concerns of the family/carers and young person, for example changes from usual behaviour
- Assess children/young people to identify likely source of infection, risk factors or indicators of clinical concern
- Assess children with particular care when communication is limited, such as where English is a second language, using aids to improve communication
- For normal age-specific values please see Royal Devon paediatric early warning score (PEWS) charts

Management of suspected paediatric sepsis:

- See "Could this be paediatric sepsis?" flow chart (page 3)
- Call for senior help early
- Antimicrobials should be prescribed as per the Royal Devon antimicrobial guidelines (or as per microbiology advice) and given within 1 hour of the time from suspected sepsis
- All decisions, treatment and timing must be documented clearly in the medical notes
- For suspected neutropenic sepsis follow Royal Devon neutropenic sepsis guideline

Could this be Paediatric Sepsis?

Any 1 of high risk criteria?

- Looks very unwell to health professional
- Non-blanching rash/ mottled/ ashen/ blue
- Reduced GCS/ altered mental state or behaviour/ change in tone or cry
- Temperature <36°C any age or ≥38°C in <3 months old or immunocompromised
- Unexplained raised respiratory rate, apnoea, grunting, oxygen sats <90%
- Persistent or unexplained tachycardia
- Hypotension or heart rate <60/min[1]

YES

NO

High risk of sepsis
Urgent Action

1. Monitoring
2. Oxygen
3. Urgent senior review (ST4+) to assess need for the following:
   - IV/IO access
   - Bloods/culture/gas with lactate
   - IV/IO antibiotics (within 1 hour)
   - IV/IO fluids
4. Consultant review
5. ITU opinion / inotrope support as deemed necessary

Any 1 of moderate risk criteria?

- Cold extremities or leg pain or leg ache
- Pallor of lips/skin/tongue
- Rigors or temperature >39°C
- Unexplained oxygen saturations <92% in air
- Central capillary refill **≥ 3 seconds**
- Significantly reduced urine output
- Unusual behaviour
- Central line or invasive procedure/ surgery/ trauma in the last 6 weeks
- Immunocompromised

YES

NO

Lower risk of sepsis

1. Clinician review
2. Manage as per diagnosis
3. Regular PEWS
4. Reconsider sepsis if deteriorates
5. Safety net before discharge

- Use PEWS charts for normal ranges and follow PEWS escalation plan.
- Follow paediatric sepsis guideline
- Follow antimicrobial guidelines.
- Use WATCh drug sheet and BCH guideline on septic shock in children.

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1. ASSESSMENT FOR PAEDIATRIC SEPSIS: ASSESS FOR HIGH/ MODERATE/ LOW RISK CRITERIA FOR SEPSIS- SEE FLOW CHART

Below gives the background and further detail for the assessment aspects on the "Could this be paediatric sepsis?" flow chart, page 3 and all risk criteria for paediatric sepsis are summarised in the Bristol Children's Hospital flowchart in Appendix I (page 15).

Table 1. High risk criteria for sepsis

Key points	Explanatory notes
Looks very unwell to health professional	
Non-blanching rash/ mottled/ ashen/ blue	
Reduced GCS/ altered mental state or behaviour/ change in tone or cry	Use GCS or AVPU scales Examples • No response to social cues • Does not wake or if roused does not stay awake • Weak high-pitched or continuous cry
Temperature <36℃ any age or ≥38℃ in under 3 months old or immunocompromised	Use axillary thermometer
Unexplained raised respiratory rate, cyanosis, apnoea, grunting or oxygen saturations <90%	• See PEWS charts for normal ranges ○ Except on 5-11 years PEWS chart, if 8-11 years old, a respiratory rate of ≥ 25/min indicates severe respiratory distress • Cyanosis of skin, lips or tongue • Low oxygen saturations: <90% in air or increased oxygen requirement over baseline with no obvious respiratory cause • Always consider sepsis in asthma and bronchiolitis as it can co-exist
Persistent or unexplained tachycardia	• For example not pain or fever related • See PEWS charts for normal ranges ○ Except on 5-11 years PEWS chart, if 8-11 years old, a heart rate of ≥ 115/min indicates severe tachycardia
Hypotension	Hypotension (as per PEWS chart): <3 months: <60 mmHg 4-11 months: <70 mmHg 1-4 years: <80 mmHg 5-11 years: <90 mmHg ≥12 years: <100 mmHg Notes: • Hypotension is a late sign of sepsis in children • Look carefully for other signs of shock before hypotension develops • Use appropriate size cuff

Table 2. Moderate risk criteria for paediatric sepsis

Key points	Explanatory notes
Cold extremities or leg pain or leg ache	
Pallor of lips, tongue or skin	
Rigors or temperature >39℃	Use axillary thermometer
Unexplained oxygen saturations <92% in air or mild increase in work of breathing	
Central capillary refill ≥3 seconds	
Significantly reduced urine output	Not passed urine in the last 12- 18 hours (or <1 ml/Kg/hour in catheterised patients <12 years old and <0.5 ml/Kg/hour in ≥12 years old)
Unusual behaviour	Examples • Not responding normally to social cues • Decreased activity • No smile • Wakes only with prolonged stimulation • Decreased activity • Parent/ carer concern that different behaviour than normal
Central line or invasive procedure/ surgery/ trauma in the last 6 weeks	
Immunocompromised	Impaired immunity from: • Treatment e.g. chemotherapy, immunosuppressant drugs • Immune deficiency e.g. asplenia, primary immune deficiency

Table 3. Low risk criteria for paediatric sepsis

Responds normally to social cues
Content or smiles
Stays awake or awakens quickly
Strong normal cry or not crying
Normal colour
No high or moderate risk criteria

2. MANAGEMENT OF SUSPECTED PAEDIATRIC SEPSIS- SEE FLOW CHART IN APPENDIX 1 AND SUMMARY DIAGRAM ON PAGE 3.

Below gives the background and further detail for the management aspects on "Could this be paediatric sepsis?" flow chart on page 3 of this guideline.

Table 5. Summary of management of suspected paediatric sepsis

Key points	Explanatory notes
Continuous monitoring (or minimum observations every 30 minutes, depending on setting)	HR, RR, oxygen saturations, BP, temperature, GCS/AVPU
Oxygen	Support airway and breathing Apply high flow oxygen if: • Shock • Oxygen saturations <92% in air • At higher oxygen saturations if clinically indicated
Urgent senior review (ST4+) to assess need for the following:	Emergency department or paediatric ST4+ • Immediate review if 1+ high risk criteria • Review within the hour if 2+ moderate risk criteria
• IV/IO access and culture/gas/bloods	Bloods • Blood culture prior to antibiotic administration if does not substantially delay treatment • Gas including glucose and lactate • Other bloods: full blood count, creatinine, urea and electrolytes, CRP, coagulation • Consider: magnesium, group and save

• IV/IO antibiotics (within 1 hour of suspected sepsis)	Give a broad-spectrum antimicrobial at the maximum recommended dose, as per Royal Devon antimicrobial guidelines
	• Usually IV ceftriaxone 80mg/Kg (maximum dose 4 grams) for suspected community acquired sepsis, or use alternative, as per Royal Devon antimicrobial guideline (or microbiology advice) Royal Devon Antimicrobial Guidelines if :
	  ○ Neonatal sepsis (i.e. less than 28 days of age) Late Onset Neonatal Sepsis
	Neutropenic sepsis Neutropenic Sepsis Guideline
	  ○ Known contraindication to ceftriaxone e.g. patient requiring a calcium infusion; cephalosporin allergy
	  ○ Previous ceftriaxone resistant organism or risk factors for ceftriaxone resistant organism
	For infants <3 months with a fever follow NICE guidance in <5s: Parenteral antibiotics for
	• <1 month old with fever

	• 1-3 months old with fever and appears unwell or abnormal white count (<5, >15 x 10^9/L) Add in amoxicillin for infants <1 month old in addition to cephalosporin to cover Listeria. Consider need for antiviral such as acyclovir to cover Herpes Simplex in encephalitis and neonatal/infant sepsis/meningitis Follow NICE antimicrobial stewardship guidelines including antibiotic review within 48-72 hours
• IV/IO fluid bolus	**Lactate (mmol/L) monitoring is useful** >4: Consider IV fluid bolus without delay and discuss with ITU 2-4: Consider IV fluid bolus without delay <2: Consider IV fluid bolus **If systolic BP <90 mmHg in >12 yrs**: IV fluid bolus without delay and discuss with ITU **Volume and choice of fluids** First choice for IV fluid bolus is Plasma-Lyte or 0.9% saline according to what is available Give 10 ml/Kg fluid bolus then reassess. Up to 40 to 60 ml/Kg IV fluid boluses may be required in the first hour titrated to clinical markers of cardiac output and discontinued if signs of fluid overload develop. Be more cautious in certain conditions with increased risk of fluid overload such as cardiac disease, renal disease, neonates, diabetic ketoacidosis May consider 4.5% human albumin solution for septic shock (available from blood transfusion) May hold fluids if haemodynamically stable and lactate <2 unless clinical state dictates otherwise If hepatomegaly or crackles on chest use inotropic support rather than fluid resuscitation **Reassessment after each bolus is essential**
4. Discuss with consultant (if not yet aware); must attend if failure to respond	Use SBAR communication tool Markers of failure to respond to initial treatment • No improvement after 2nd fluid bolus • Systolic BP persistently below the 5th centile for age • Reduced level of consciousness despite resuscitation • High lactate not reduced by more than 20% of initial value in the first hour (or remains >2 after 1 hour) • Need for mechanical ventilation
5. ITU review/ inotropic support	ITU review to consider need for central venous access and initiation of inotropes or vasopressors and need for mechanical ventilation

6. Further management in shock	See WATCh guideline on sepsis See BCH guideline on Septic Shock in children See PICU drug sheet Anticipate, monitor and correct • Metabolic abnormalities: hypoglycaemia, hypocalcaemia, hypomagnesaemia, metabolic acidosis • Anaemia and bleeding disorders In refractory shock evaluate for and treat if present • Pneumothorax • Pericardial tamponade • Endocrine emergencies
7. Further management following improvement	Treat as per diagnosis • Follow diagnosis specific guidelines (NICE, Royal Devon) If no definitive condition identified • Repeat structured assessment for sepsis hourly • See Table 8 for targeted investigations

Consider treatment adjuncts in specific types of sepsis

Meningitis- Dexamethasone 150 mcg/Kg qds- see Trust paediatric antimicrobial guidelines

meningitis section for criteria for dexamethasone

Group A streptococcal toxic shock syndrome- Clindamycin; IV immunoglobulin

Viral sepsis- aciclovir

Table 6. Summary of management of lower risk of paediatric sepsis

Key points	Explanatory notes
Clinician review	• Repeat structured assessment for sepsis hourly • If no high or moderate risk criteria ensure senior decision maker review within 3 hours for consideration of antibiotics • See table 7 for targeted investigations
Manage as per diagnosis	• See associated Royal Devon guidance and NICE guidance relevant to diagnosis
Regular PEWS observations	Frequency of observations according to clinical need
Reconsider sepsis if deteriorates	
Safety net on discharge	Give appropriate safety netting advice and leaflet Further recommended parent information- see UK sepsis trust

3. IDENTIFICATION OF SOURCE OF INFECTION

3.1 History

- Perform though history to help identify source of infection
- Ensure all risk factors for sepsis have been considered, and consider possible sources arising from any risk factors

Table 7. Risk factors for sepsis

Age < 1 years old
Infant born with risk factors for infection (preterm birth; maternal or previous infant with Group B streptococcus; maternal intrapartum fever/ chorioamnionitis; maternal prolonged rupture of membranes ≥ 24 hours)
Recent trauma/ surgery/ invasive procedures (within the last 6 weeks)
Impaired immunity (from treatment such as chemotherapy, immunosuppressant drugs or from immune deficiency (asplenia, primary immune deficiency, Diabetes Mellitus)
Any breach of skin integrity (cut, burn, blister, skin infection, indwelling line or catheter)
Young women who are or have been pregnant in the last 6 weeks or tampon use

3.2 Examination

Perform a thorough systemic examination to identify source of infection, including for sources that might need surgical drainage.

3.3 Sepsis Mimics

Consider causes of immune dysregulation and sepsis mimics including:

- Paediatric Inflammatory Multisystem Syndrome (PIMS-TS)- defined by evidence of inflammation (persistent fever, neutrophilia, elevated CRP, lymphopaenia) and single/multi-organ failure with known or suspected temporal association with prior Covid-19 infection without other identifiable microbiological cause. PIMS-TS should be managed in line with Bristol PIMS-TS guidance: https://royaldevonstaff.nhs.uk/download.cfm?ver=12619

- Haemophagocytic Lymphohistiocytosis (HLH)- defined as a syndrome of severe, dysregulated inflammation which can lead to cytokine storm and multi-organ failure. It should be considered in children with persistent fever, coagulopathy and thrombocytopaenia/pancytopaenia. A serum Ferritin >10,000 µg/L is both sensitive and specific. For further details refer to Bristol Children's Hospital Septic Shock in Children: Management in PICU, page 8: https://royaldevonstaff.nhs.uk/download.cfm?ver=8205

Particularly in infants other sepsis mimics include:

- Cardiac causes of collapse- including duct dependent lesions, non-duct dependent lesion, cardiac arrhythmias and cardiomyopathies. Congenital cardiac disease that is dependent on a patent ductus arteriosus affects 6-8 per 1000 live births. It should be considered in infants with cyanosis, hypoxia in spite of oxygen therapy and tachypnoea in the absence of clear lung pathology.
- Inborn errors of metabolism which may present in infants with poor feeding, vomiting and lethargy. It should be considered where there was an uncomplicated birth followed by acute deterioration. There may be a history of consanguinuity, previous unexplained sibling mortality or recurrent miscarriages/intrauterine death, abnormal odour.

3.4 Investigations

Tailor investigations for source depending on findings on history and examination.

Table 8. Targeted investigations to identify source of infection

Specific age range/ suspected source of infection	Useful investigations
All cases	Consider • CXR • Urine dipstick and culture • If no source found consider investigations for a hidden source such as abdominal/pelvis imaging

Neonates	Blood culture Urine dipstick and culture Lumbar puncture if no contraindication
Infant under 3 months who appear unwell/ have an abnormal white count (<5 or >15 x10^9/L)	Blood culture Urine culture Lumbar puncture if no contraindication
Septicaemia	Blood culture Blood for pneumococcal and meningococcal PCR
Pneumonia	Blood culture Viral nose swab CXR
Pyelonephritis	Blood culture Urine dipstick and culture Consider renal tract ultrasound
Meningitis	Blood culture Blood for pneumococcal and meningococcal PCR Lumbar puncture for MCS, virology, biochemistry- perform only when safe and when no contraindications (see Table 8)
Bone/joint infection	Blood culture Imaging Surgical opinion
Skin infection	Skin swab for MCS
Abdominal/pelvic infection	Blood culture Imaging Surgical and gynaecological opinions

Table 9. Contraindications to lumbar puncture

Superficial infection at LP site
Respiratory insufficiency
Shock
Extensive or spreading purpura
Coagulation abnormalities
If convulsions until stable
Reduced or fluctuating level of consciousness (GCS<9 or drop of ≥3)
Signs suggesting raised intracranial pressure
Relative bradycardia and hypertension
Focal neurological signs
Abnormal posture or posturing
Unequal, dilated or poorly responsive pupils
Papilloedema
Abnormal dolls' eye movements

4. INFORMATION AND SUPPORT FOR CHILDREN AND FAMILIES/ CARERS

4.1 Information for children and families/ carers when diagnosed with sepsis

- Nominate a team member to give information to children and families/ carers, particularly in the emergency department, then later on the ward, on
    - Explanation of sepsis
    - Investigations and management plan
    - Regular timely updates on treatment, care and progress
- Key points
    - Avoid medical jargon
    - Give opportunities for questions to be asked and answered, repeating when needed
    - Give information on national charities and support groups for sepsis such as the UK sepsis trust- https://sepsistrust.org; and the Surviving Sepsis Campaign- http://www.survivingsepsis.org

4.2 Information for children and families/ carers on discharge

- When sepsis has been diagnosed
    - Ensure family and GP are aware of diagnosis and what it means
    - Give people opportunity to discuss concerns such as why they developed sepsis, whether they are more likely to get it again, if more investigations are necessary, details of community care (e.g. use of PICC lines), what to expect during recovery, follow up arrangements and complications
    - Give information on national charities and support groups for sepsis
    - Give appropriate safety netting advice, as below
- When sepsis suspected but not diagnosed or in people at risk of sepsis (e.g following surgery or chemotherapy)

- Give information on what is sepsis, why it was suspected, investigations and treatment given
- Give safety netting advice, and ensure it has been understood
  - Advise what symptoms to monitor for and when and how to seek medical attention
  - How to get medical attention urgently if needed

- Sources of information
  - Royal Devon patient leaflet on fever in under 5s
  - HANDi Paediatric App: https://www.royaldevon.nhs.uk/stay-well/handi-children-and-young-people-s-app/
  - National charities:
    - UK sepsis trust- https://sepsistrust.org
    - The surviving sepsis campaign- http://www.survivingsepsis.org/Pages/default.aspx

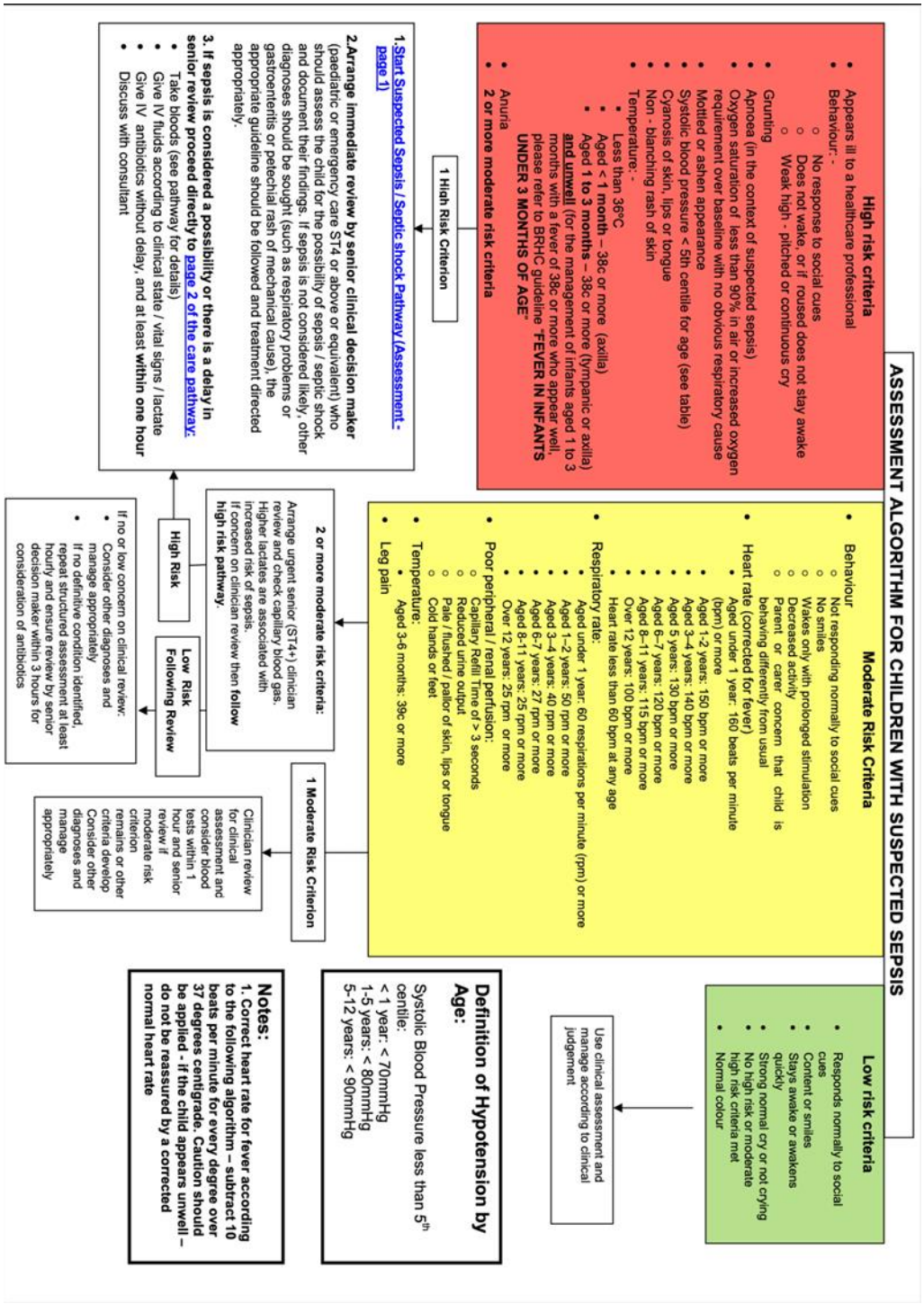
5. ROYAL DEVON ASSOCIATED DOCUMENTS

- Royal Devon PEWS charts for age-specific values
- Royal Devon antimicrobial guidelines:
- Royal Devon fever in children aged 3 months-5 years guidelines
- Royal Devon fever in children under 3 months
- Royal Devon neutropenic sepsis guideline
- Royal Devon patient leaflet on fever in under 5s
- Royal Devon initial investigation of acute presentations of suspected Inborn Errors of Metabolism (IEM) guideline

6. REFERENCES

- NICE guideline 51 on Sepsis: recognition, diagnosis and early management, published in July 2016, last updated 2017
- NICE guideline 143 on Fever in under 5s: assessment and initial management, published 2019, last updated 2021
- NICE guideline 102 on Meningitis and Meningococcal septicaemia in under 16s: recognition, diagnosis and management, published 2010, last updated 2015
- Surviving Sepsis Campaign Guidelines, *Critical Care Medicine*, published October 2021.
- Bristol Children's Hospital guideline: Sepsis and suspected sepsis- management and care pathway for the children's emergency department, last updated 2020
- Bristol Children's Hospital guideline: Septic shock in children: Management on PICU, last updated 2021
- The Leeds Teaching Hospitals NHS Trust: Considering sepsis in paediatrics
- Surviving sepsis campaign: special considerations in children
- Meningitis research foundation: Management of meningococcal disease in children and young people
- Bristol Children's Hospital guideline: Paediatric Multisystem Inflammatory Syndrome temporally associated with COVID-19 (PIMS-TS) guideline, last updated October 2021
- Lal, N. and Varshney, T. The collapsed newborn in the emergency department. *BJA Education* 2018; 18(8): 254-258. doi:10.1016.j.bjae.2018.05.004

APPENDIX 1

Bristol Children's Hospital Assessment Algorithm for Children with Suspected Sepsis

(From: Sepsis and Suspected Sepsis – Management Guideline and Care Pathway for the Children's Emergency Department http://foi.avon.nhs.uk/download.aspx?did=20191)